Lead 16.2: a fast 4E test as a subgroup check

Round 4 of `RESEARCH_ISOGENY_FIBERS_ECDLP.md` (§14.2, §16.2) · 2026-10-11 · worktree commit `440c0e163` + untracked round-4 files

Code: `examples/r4_16_2_fast_4e.rs` · runs started 2026-10-11T19:50:38Z · x86_64, single thread, contended host

1. Question and verdict

On the a = 0 Koblitz curve $y^2 + xy = x^3 + 1$ over $\mathbb{F}_{2^n}$, n odd, with $\#E = 4r$, does the closed form of Prop. 14.2 decide membership in $G = 4E$ exactly, and what does it cost against $[r]P = O$? The falsifier is any disagreement.

Verdict: confirmed (measured).

- 0 errors on every point for $n = 5, 7, \ldots, 23$ (exhaustive).
- 0 errors on 10^6 sampled points each at $n = 131$ (the ECC2K-130 curve) and $n = 233$ (sect233k1), across all four cosets of $4E$.
- On members of G the test costs 1 M, 3 traces, and either $n - 1$ S or $\lceil n/8 \rceil$ table lookups. τ-NAF $[r]P$ costs 346 M + 609 S ($n = 131$) and 618 M + 1085 S ($n = 233$).
- **No novelty is claimed.** A two-level halving validation for cofactor 4 already appears in the literature (§5).

2. The test (derived)

Doubling (u, v) gives $\lambda = u + v/u$, $x = \lambda^2 + \lambda$ and $y = u^2 + (\lambda + 1)x$.

- P has a rational half iff $\mathrm{Tr}(x) = 0$. For odd n, $\lambda = H(x)$ or $H(x) + 1$, where H is the half-trace.
- A half Q lies in $2E$ iff $\mathrm{Tr}(u) = 0$. Since $\mathrm{Tr}(u) = \mathrm{Tr}(u^2) = \mathrm{Tr}(y) + \mathrm{Tr}(\lambda x) + \mathrm{Tr}(x)$, this is $\mathrm{Tr}(y) + \mathrm{Tr}(xH(x))$ when $\mathrm{Tr}(x) = 0$.
- The two choices of λ give Q and $Q + T$. They agree because $T = 2 \cdot (1, 0) \in 2E$.

> $P \in 4E \iff P = O$ or $[\mathrm{Tr}(x) = 0$ and $\mathrm{Tr}(y) + \mathrm{Tr}(x \cdot H(x)) = 0]$. The traced elements are x, y and the product $x \cdot H(x)$.

- $T = (0, 1)$ is rejected automatically, since $\mathrm{Tr}(1) = n \bmod 2 = 1$.
- $(1, 0)$ has order 4 and no half, because $\mathrm{Tr}(1) = 1$. So the 2-part is exactly $\mathbb{Z}/4$ for every odd n, and $4E = G$ when r is prime.
- Preconditions: P is on the curve, n is odd and $a = 0$.

3. Measured results

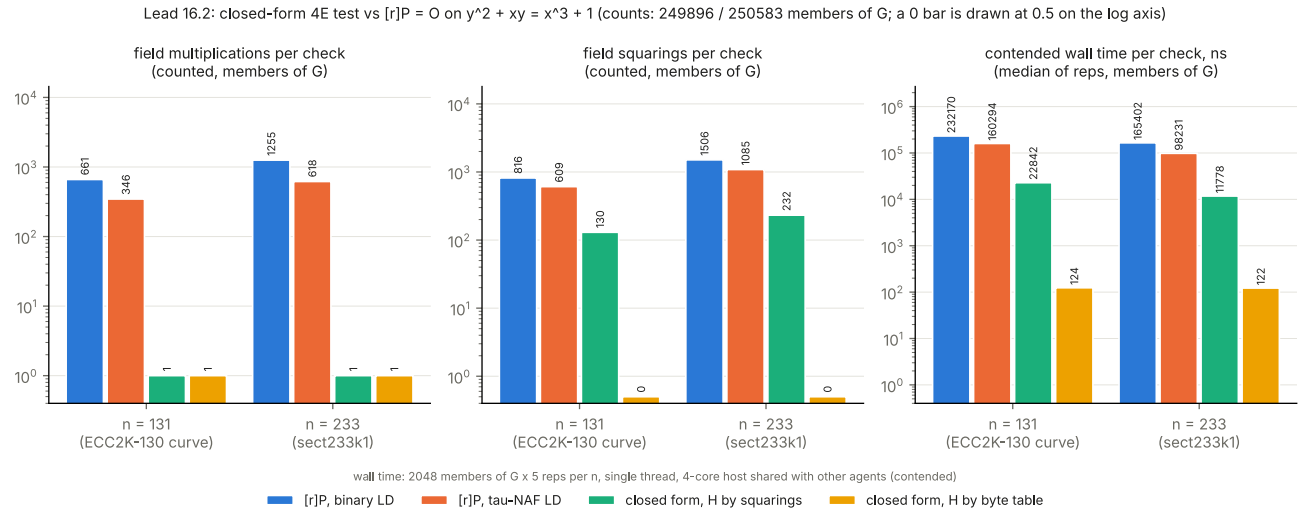


Figure 1: Counted field operations (left, middle) and contended wall time (right) per check, for members of G. Counts come from 249 896 ($n = 131$) and 250 583 ($n = 233$) points. Wall time is the median of 5 reps over 2048 points. Sources: `sample_n131.json`, `sample_n233.json`, `plot_data.json`.

n	true coset	in G	accept	reject	bits (Tr x, Tr y + Tr xH)	errors
131	0	yes	249 897	0	(0,0)	0
131	1	no	0	249 504	(1,0)	0
131	2	no	0	250 049	(0,1)	0
131	3	no	0	250 559	(1,1)	0
233	0	yes	250 584	0	(0,0)	0

n	true coset	in G	accept	reject	bits (Tr x, Tr y + Tr xH)	errors
233	1	no	0	249 533	(1,1)	0
233	2	no	0	249 993	(0,1)	0
233	3	no	0	249 899	(1,0)	0

Exhaustive runs found 0 false accepts and 0 false rejects at every odd n from 5 to 23; the largest is $n = 23$ with 8 383 412 points.

Further checks:

- Binary and τ-NAF $[r]P$ agree on every point.
- The library's `is_valid_public_point` agrees on 2000 of 2000 points per n.
- $r(131)$ equals the literature ECC2K-130 order, cross-checked by search only. $r(233)$ equals the library's sect233k1 order.

Observation (not derived): on these samples the bit pair also separates cosets 1 and 3, with the assignment swapped between $n = 131$ and $n = 233$.

n	method	M	S	traces / lookups	contended wall, median
131	$[r]P$ binary LD	661	816	0	232 µs
131	$[r]P$ τ-NAF LD	346	609	0	160 µs
131	closed form, H by squarings	1	130	3 / 0	22.8 µs
131	closed form, H by table	1	0	3 / 17	0.124 µs
233	$[r]P$ binary LD	1255	1506	0	165 µs
233	$[r]P$ τ-NAF LD	618	1085	0	98 µs
233	closed form, H by squarings	1	232	3 / 0	11.8 µs
233	closed form, H by table	1	0	3 / 30	0.122 µs

The counts match the derived formulas: binary $3(L-1) + 8(w-2) + 2$ M; τ-NAF $8(z-2) + 2$ M; the closed form 1 M + $(n-1)$ S. Multiplication ratio, τ-NAF against the closed form: 346× and 618×. Wall-time ratio, τ-NAF against the table variant: about 1300× and 800×. These are contended, with untuned squaring, so they are order-of-magnitude only.

4. Limits

- The exhaustive check stops at $n = 23$. A run to $n = 25$ was stopped to respect the per-run time limit.
- Not constant-time. Odd n and $a = 0$ only.
- There is no `identity.certificate/v1` record for the identity.
- `cargo clippy -D warnings` fails on 28 errors in `src/` at this commit. The example itself has no lints. `rustfmt --check` passes.

5. Literature status (search snippets only; PDFs blocked)

- Halvability $\mathrm{Tr}(x) = \mathrm{Tr}(a)$: Knudsen 1999; Schroeppel; Fong–Hankerson–López–Menezes.
- US patent 7,257,709 / US 2004/0114760 (priority 2002): validation of binary-curve keys with cofactor 2 or 4 by testing for a half via traces.
- King, PKC 2004 (LNCS 2947).
- Eagle–Galbraith–Ong, ePrint 2009/086. For $\#E = 4\ell$: halve, then check $\mathrm{Tr}(x_Q) = \mathrm{Tr}(a^2)$.
- Pornin, ePrint 2022/1164: membership by halvings on twisted Edwards curves.
- ePrint 2026/749 (Divide-and-Pair).

The closed form here only inlines the half's trace, so no square root or second half-trace is needed. No novelty is claimed.